inception in the late 1600s until the beginning of this century, Newton's theory of gravity passed this test with flying colors. Whether applied to balls thrown up in the air, objects dropped from leaning towers, comets whirling around the sun, or planets going about their solar orbits, Newton's theory provides extremely accurate explanations of all observations as well as predictions that have been verified innumerable times in a wealth of situations. The motivation for questioning this experimentally successful theory, as we have emphasized, was its property of instantaneous transmission of the gravitational force, in conflict with special relativity.

The effects of special relativity, although central to a fundamental understanding of space, time, and motion, are extremely small in the slow-velocity world we typically inhabit. Similarly, the deviations between Einstein's general relativity—a theory of gravity compatible with special relativity—and Newton's theory of gravity are also extremely small in most common situations. This is both good and bad. It is good because any theory purporting to supplant Newton's theory of gravity had better closely agree with it when applied in those arenas in which Newton's theory has been experimentally verified. It is bad because it makes it difficult to adjudicate between the two theories experimentally. Distinguishing between Newton's and Einstein's theories requires extremely precise measurements applied to experiments that are very sensitive to the ways in which the two theories differ. If you throw a baseball, Newtonian and Einsteinian gravity can be used to predict where it will land, and the answers will be different, but the differences will be so slight that they are generally beyond our capacity to detect experimentally. A more clever experiment is called for, and Einstein suggested one.10

We see stars at night, but of course they are also there during the day. We usually don't see them because their distant, pinpoint light is overwhelmed by the light emitted by the sun. During a solar eclipse, however, the moon temporarily blocks the light of the sun and distant stars become visible. Nevertheless, the presence of the sun still has an effect. Light from some of the distant stars must pass close to the sun on the way to earth. Einstein's general relativity predicts that the sun will cause the surrounding space and time to warp and such distortion will influence the path taken by the starlight. After all, the photons of